Progress Report 1

CSIS 4495 - Applied Research Project

Student: Jasleen Chhabra | Project partner: Anushka Kshirsagar | Reporting period: Oct 6 - Oct 11, 2026

Work Date/Hours Logs

Date	Hours	Description of work done
Oct 6, 2026	0.5	Reviewed the project proposal again to list the data our system needs (sales history, stock, purchase orders with ETAs, lead times, pack sizes).
Oct 7, 2026	1.5	Morning: studied public dataset websites (M5 Forecasting, UCI Online Retail II, DataCo Smart Supply Chain).
Oct 7, 2026	1.5	Evening: met with my partner Anushka, continued the review, and shortlisted the resources that are valuable for our project.
Oct 11, 2026	1.0	Met with Anushka and finalized the data we will use for the project, based on what the public datasets do and do not cover.
Oct 11, 2026	1.5	Generated the project dataset (45 SKUs, sales, inventory, purchase orders) with a Python script using AI assistance. Rechecked all the spreadsheets myself.
Oct 11, 2026	0.5	Checked the data files (dates, stock and purchase order values) and reviewed the output tables.
	Total: 6.5	

Summary Description of Work

This week, I worked on the data for our project. On Oct 6, I read our proposal again to list the data our system needs: sales history, current stock, purchase orders with multiple ETAs, lead times, and pack sizes. On Oct 7, I studied public dataset websites, including M5 Forecasting, UCI Online Retail II, and DataCo Smart Supply Chain, to see which ones fit our project. My partner Anushka and I met to decide which resources are valuable for us.

One issue was that none of these datasets has sales, purchase orders, ETAs, and lead times together. Online Retail II was the most useful for sales patterns, but it has no stock or purchase order data. So, on Oct 11, we finalized the data we will use, and I generated the project dataset (45 SKUs, monthly sales from Jan 2024 to Sep 2026, inventory, and about 295 purchase orders) with a Python script using AI assistance. I also rechecked all the spreadsheets myself.

Next week, Anushka and I will meet to decide and divide the work for the next stage of the project, and I will update my timeline after that meeting.

Repo Check-in of Implementation Completed

The files I have checked in the repo are as follows: Implementation/Data/products.csv, sales_monthly.csv, inventory.csv, purchase_orders.csv, inventory.db, generate_dataset.py, and README.txt (data dictionary). My progress report is in ReportsAndDocuments.

AI Use Section

Table of AI Tools and Specific Use

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used	Value Addition
Claude (Anthropic)	Sonnet 5.5, Free account	Generated the sample dataset and the Python generation script from our proposal's schema; suggested public dataset links; helped draft this report.	I decided what data our project needs from the proposal. I studied the public datasets myself and chose which ones are useful. I rechecked all the spreadsheets and worked with my partner to finalize the approach. I also edited this report.

Appendix: Prompt History

1. I researched three public datasets for my inventory and procurement project and shortlisted them: M5 Forecasting (https://www.kaggle.com/c/m5-forecasting-accuracy), UCI Online Retail II (https://archive.ics.uci.edu/dataset/502/online+retail+ii), and DataCo Smart Supply Chain (https://data.mendeley.com/datasets/8gx2fvg2k6/5). Please examine these pages and tell me, for each dataset, which of these fields it contains: sales history, stock on hand, purchase orders, expected arrival dates, lead times, supplier and pack size. Put the answer in a table and say clearly if you could not open a page.
2. Based on that table, explain which dataset is the most useful for sales and seasonality patterns, and what is missing for my project. I will write my own conclusion from your answer.
3. I downloaded the Online Retail II file (online_retail_II.xlsx). Write pandas code that removes cancellations (InvoiceNo starting with C) and negative quantities, then groups the data into monthly units sold per StockCode. Show me how to pick the top 40 to 50 SKUs by total sales.
4. For the top SKUs, write code to plot the monthly sales of three SKUs and show which months have seasonal peaks. Explain the code line by line.
5. Write code to convert the monthly sales into my schema: sku, warehouse, month, units_sold. Use neutral labels like SKU001 and assign warehouses WH1 and WH2.
6. I have also downloaded the DataCo file (DataCoSupplyChainDataset.csv). Write code to list its columns and tell me which of them relate to shipping days or delivery delays, so I can check whether they help with lead-time analysis.
7. I downloaded the M5 Forecasting files (sales_train_evaluation.csv, calendar.csv, sell_prices.csv). Write pandas code to select about 45 items from one store, convert the daily sales into monthly units sold, and show the first rows.
8. Using the M5 calendar file, explain how to mark holiday and event months (like Christmas and Thanksgiving) in the monthly sales, so I can see their effect on demand.
9. I have monthly sales from Online Retail II and from M5. Compare them: which one shows clearer seasonality for my 45 SKUs? Write code that calculates, for each SKU, the ratio of the peak month to the average month.
10. Write code that splits each dataset into a training part and a holdout part (the last 3 months), so that I can test a forecast later. Explain why we hold out recent months instead of random rows.

11. My project needs purchase orders with multiple ETAs per SKU, but the public datasets do not have them. List which tables and columns are missing in each dataset compared with my schema (products, sales_monthly, inventory, purchase_orders), as a table.
12. Online Retail II has missing CustomerID values and different unit prices for the same StockCode. Write code to check how many rows are affected, and explain whether this matters for monthly unit sales.
13. Write a short data dictionary (column name, type, meaning, example) for the monthly sales table I created from the public data, in a README format I can put in my GitHub repo.
14. Write a validation script for my final tables that checks missing values, duplicate keys, negative values, and SKUs that appear in one table but not another, and prints a clear pass/fail summary.
15. I will cite these datasets in my report. Give me the APA citation format for each of the three datasets, and tell me which details I still need to check myself (such as the year and the DOI).
16. Review the code you wrote in the previous answers and list its assumptions and limitations (for example, how it treats cancellations, missing months and different pack sizes), so that I can write them in my report.